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STL Vector in C++ - Detailed Presentation Notes

Introduction

A vector in C++ is a sequence container provided by the Standard Template Library (STL). It behaves like a dynamic array, meaning its size can change during runtime. Unlike static arrays, vectors automatically handle memory allocation and resizing.

Why Use Vectors?

- 1 No need to define size in advance
- 2 Automatic resizing when elements are added or removed
- 3 Efficient memory management
- 4 Easy insertion and deletion operations
- 5 Widely used in modern C++ programming

Declaration and Initialization

```
#include <vector>
using namespace std;
vector<int> v1; // empty vector
vector<int> v2(5); // vector of size 5 (default values)
vector<int> v3(5, 10); // vector of size 5 with all values = 10
```

Adding Elements

```
v1.push_back(10); // adds 10 at end
v1.push_back(20); // adds 20 at end
```

Accessing Elements

```
cout << v1[0]; // direct access
```

```
cout << v1.at(1); // safe access (throws error if out of range)
```

Traversing a Vector

```
for(int i = 0; i < v1.size(); i++) {  
    cout << v1[i] << endl;  
}
```

Important Member Functions

1 size() -> returns number of elements

2

capacity() -> returns allocated storage

push_back() -> adds element at end

4

pop_back() -> removes last element

5

clear() -> removes all elements

6

empty() -> checks if vector is empty

Complete Example Program

```
#include <iostream>  
#include <vector>  
using namespace std;  
int main() {  
    vector<int> v;  
    // inserting elements  
    v.push_back(10);  
    v.push_back(20);  
    v.push_back(30);  
    // displaying elements  
    for(int i = 0; i < v.size(); i++) {  
        cout << v[i] << " ";  
    }  
    return 0;  
}
```

Advantages of Vectors

1 Dynamic resizing

2

Safe and easy to use

3

Efficient memory usage

4

Built-in functions simplify coding

Disadvantages

1 Slightly slower than arrays due to overhead

2

Reallocation may occur when resizing

Conclusion

Vectors are one of the most important containers in C++. They provide flexibility, safety, and ease of use. Understanding vectors is essential for mastering modern C++ programming.